PanSt3R: Multi-view Consistent Panoptic Segmentation

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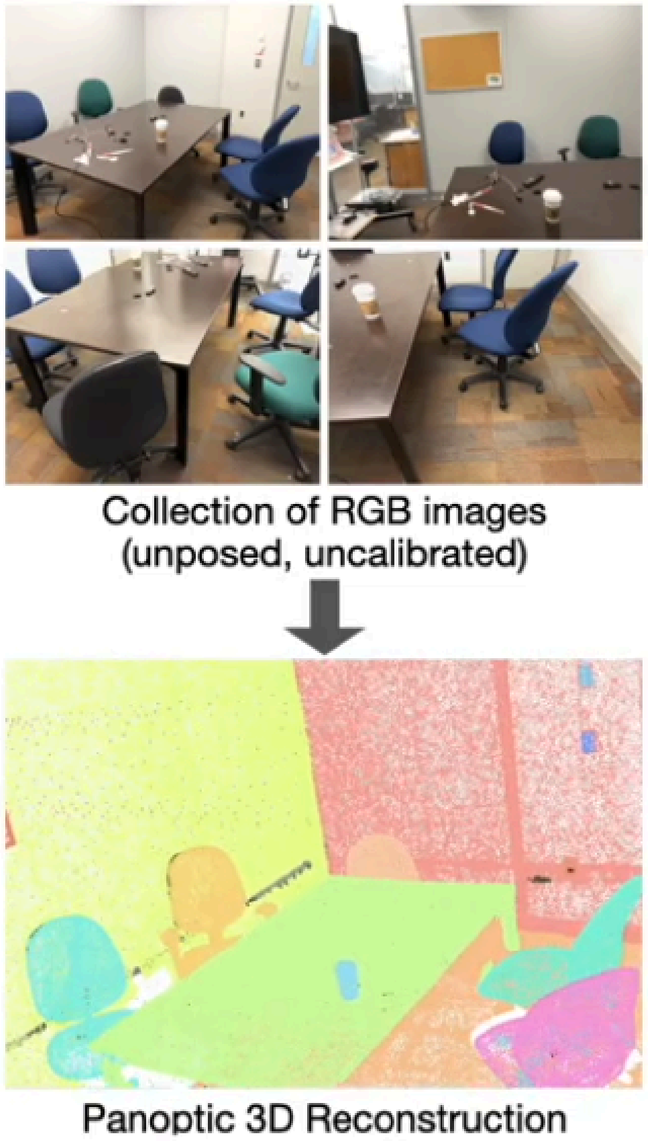
[1] Lojze Žust et al. 'PanSt3R: Multi-view Consistent Panoptic Segmentation', ICCV 2025

Summary of Motivation [1]

- Current approaches for 3D panoptic scene segmentation:
  - Require specific inputs: **point cloud**, **posed RGB images**, etc
  - Require **costly test-time optimization**
    - For aligning 2D segmentations with 3D geometry
  - Separate the 2D segmentation and 3D reconstruction pipelines
    - potentially **sacrificing efficiency**, **speed** and **accuracy**
- 3D foundation models like **MUSt3R**, **VGGT,** etc which enable fast modelling of the 3D geometry

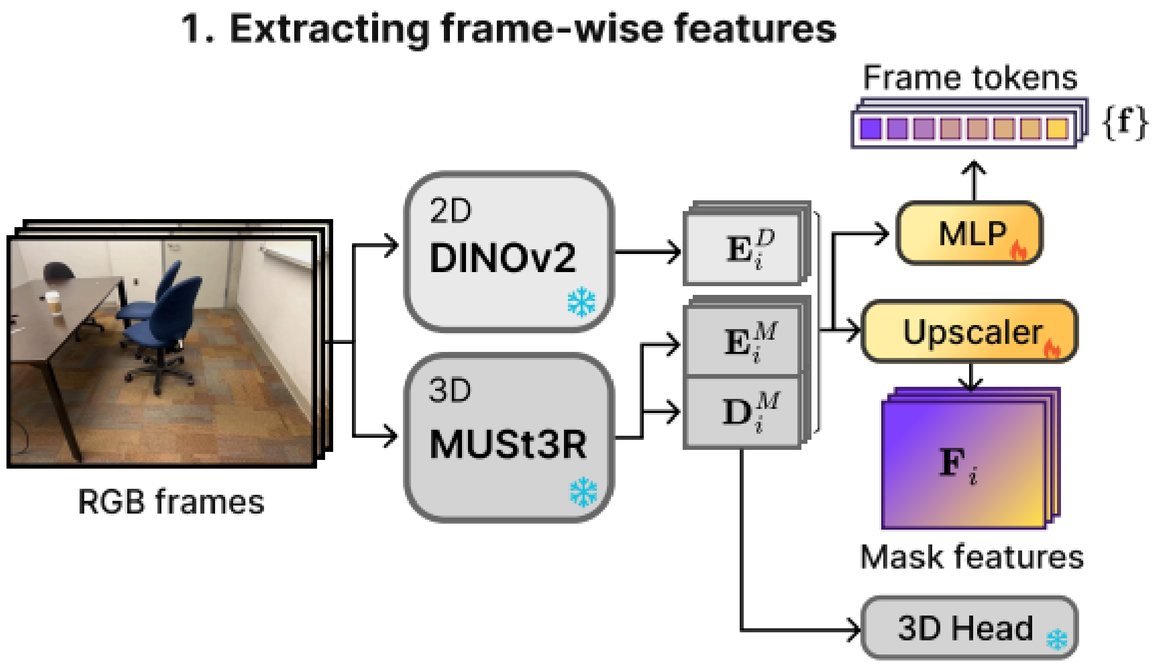
- The authors motivation was to investigate whether such models can be enhanced with **panoptic segmentation** in a more unified approach



[1] Lojze Žust et al. 'PanSt3R: Multi-view Consistent Panoptic Segmentation', ICCV 2025, https://www.youtube.com/watch?v=f1YY1XJJ-HA

PanSt3R Architecture [1]

Step 1: Extracting 2D & 3D Features



Harnesses the capabilities of 2 pre-trained feature extractors:

- **MUSt3R** extracts 3D-aware features
- **DINOv2** extracts dense semantic 2D features

These features constitute the input to:

- An MLP which produces the frame tokens
- An Upscaler producing high-res mask features (concatenated feature maps → series of MLP and 2X upsampling operations)
- A 3D Head which generates per-frame pointmaps

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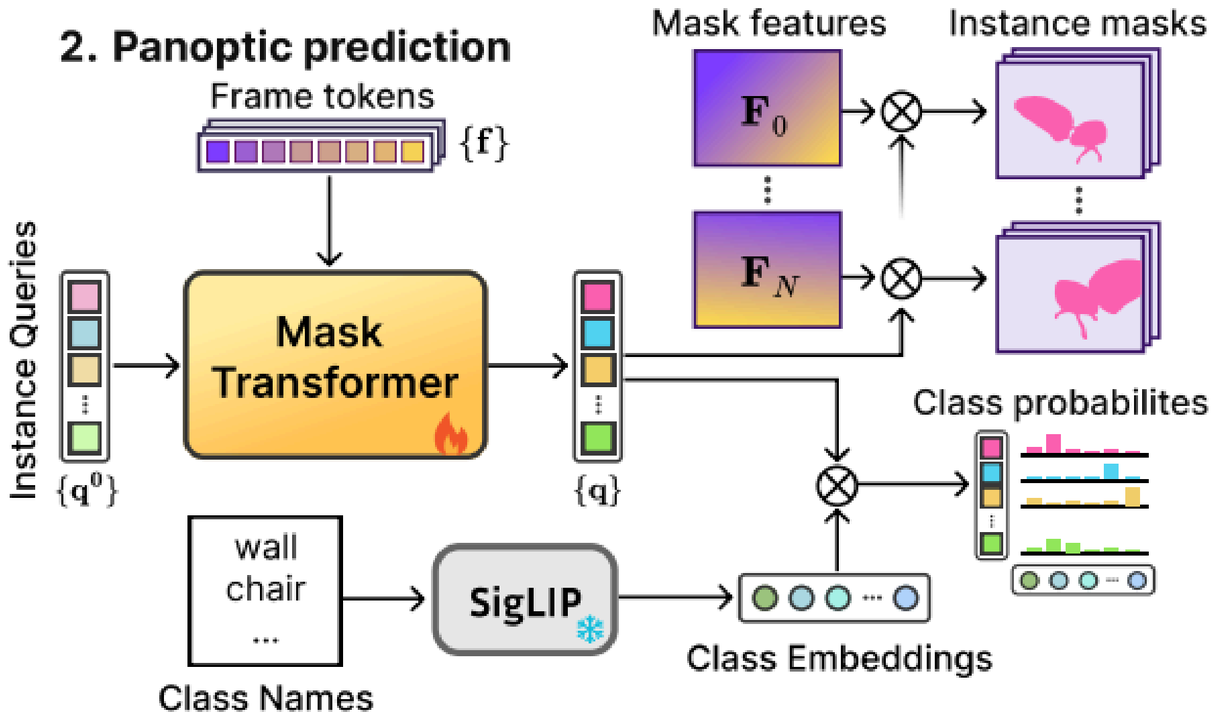
PanSt3R Architecture [1]

Step 2: Multi-view consistent panoptic predictions



Learnable Instance Queries:

- A series of learnable query features
- Every query targets one 3D object instance in the scene

Mask Transformer Decoder:

- Attends to multi-view frame tokens using cross-attention

Prediction of Instance Masks:

- Dot product of instance query and high-resolution mask features

Prediction of Instance Classes:

- Cosine similarity between the query embeddings and SigLIP generated text embeddings of class names

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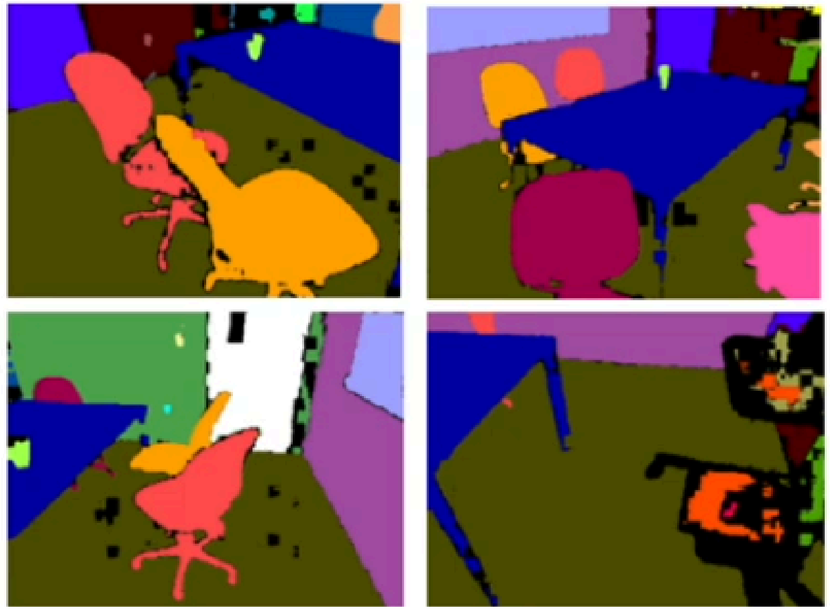
Mask Merging [1]

Improved using QUBO

Now we need to merge instance class and mask predictions for each query

Traditional Merging - Mask Transformer

Filtering out low-confidence predictions, performs well for single images but for multi-view it can lead to parts of image remaining unlabelled



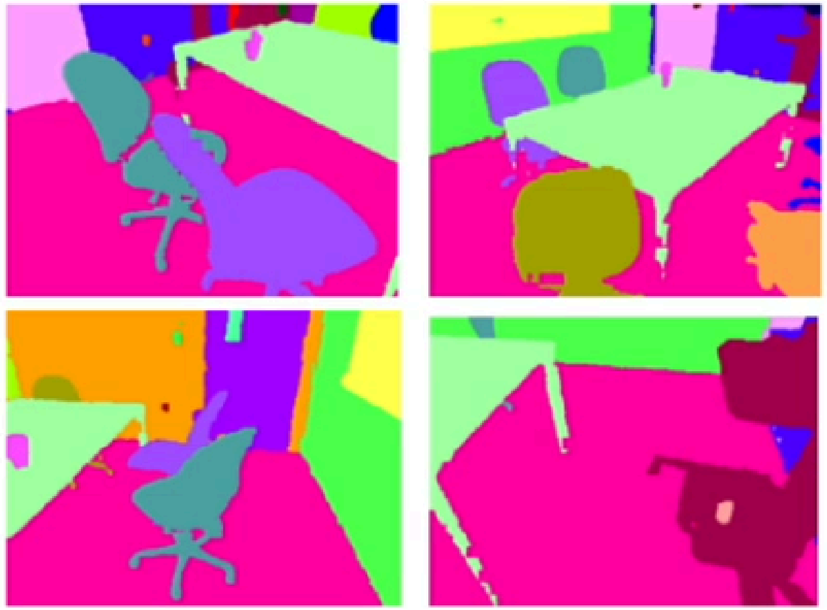
Classical Post-Processing

PanSt3R solution: Optimization using **QUBO** to select instances

Optimal **maximum coverage**, **Minimize overlap** between selected masks

Mathematically can be formalized as quadratic unconstrained binary optimization (QUBO) problem:

$$u^\star = \max_{u \in \{0,1\}^m} \sum_i u_i Q_i - \sum_{i<j} u_i u_j Q_{i,j},$$



Using QUBO

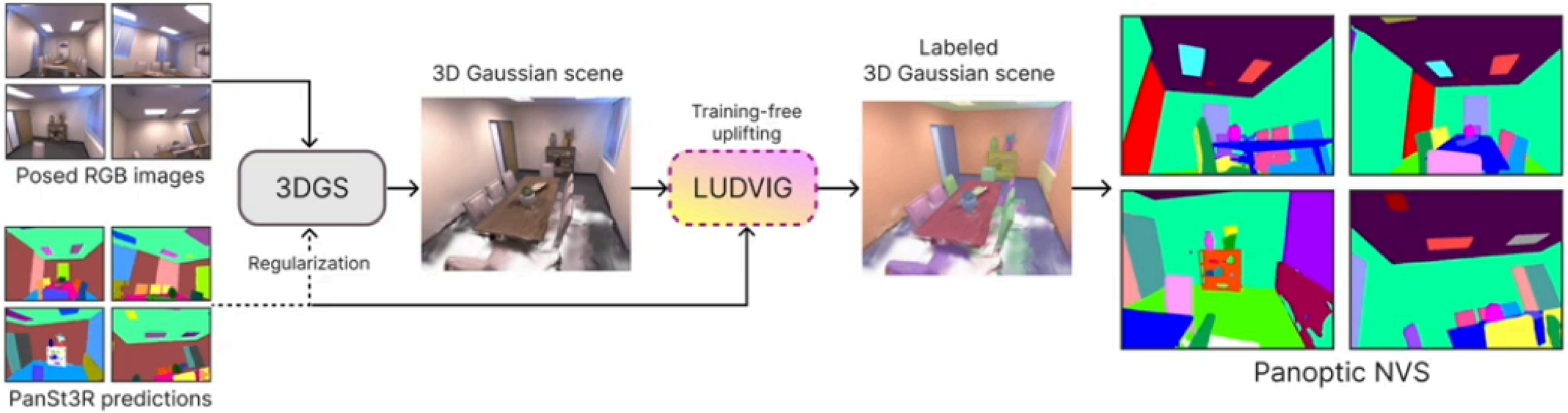
[1] Lojze Žust et al. ‘PanSt3R: Multi-view Consistent Panoptic Segmentation’, ICCV 2025, https://www.youtube.com/watch?v=f1YY1XJJ-HA

Panoptic NVS with LUDVIG [1] [2]

LUDVIG: 3DGS-based **feature uplifting** method

- simply averages 2D pixel features across all views

PanSt3R approach: Don't use it to uplift features, instead use it to it to **uplift one-hot encoded instance labels** into the optimized Gaussian Splatting scene



[1] Lojze Žust et al. 'PanSt3R: Multi-view Consistent Panoptic Segmentation', ICCV 2025, https://www.youtube.com/watch?v=f1YY1XJJ-HA

[2] Juliette Marrie et al. 'LUDVIG: Learning-free Uplifting of 2D visual features to Gaussian Splatting scenes', ICCV 2025

Experiment Setup and Evaluation Metrics

01

PanSt3R utilized a **diverse mix of 2D and 3D datasets** during training to improve generalization and robustness.

Table 1. Statistics for datasets used during training (top) and evaluation (bottom). *"MV"* denotes available multi-view data.

	Dataset	real	MV	# scenes	# classes
Training	**ScanNet++** [69]	✓	✓	855	100
	ASE [1]	x	✓	750	44
	Infinigen [44]	x	✓	936	76
	COCO [32]	✓	x	118k	80
	ADE20k [76]	✓	x	20k	150
Eval	**ScanNet++** [69]	✓	✓	50	100
	ScanNet [12]	✓	✓	12	20
	Hypersim [46]	x	✓	6	20
	Replica [53]	x	✓	7	20

02

Evaluation was conducted primarily on two benchmarks:

- **Panoptic Lifting (PanLift) Benchmark:** Used to compare performance on novel, unseen views, utilizing scenes from **ScanNet** (12 scenes), **Hypersim** (6 scenes), and **Replica** (7 scenes). These evaluations typically use 20 fixed classes.
- **ScanNet++ Validation Set:** Used for complex scenes with a **much larger number of classes (100)**, requiring more fine-grained segmentation.

PanSt3R is evaluated against existing methods (which typically require camera poses and test-time optimization) by generating panoptic predictions for **novel viewpoints.**

03

The primary metric used is **Panoptic Quality (PQ)**, which measures both segmentation quality and recognition quality.

1. **Scene-PQ (PQsc):** PQ computed at the scene level by concatenating predictions across all images, ensuring predictions are tied between all views. The paper uses PQ to denote scene-PQ.
2. **PQth and PQst:** Panoptic quality computed separately for thing classes (countable objects) and stuff classes (uncountable concepts)

Panoptic Quality (PQ). The Panoptic Quality (PQ) score [24] is defined as

$$\mathrm{PQ} = \frac{2\sum_{(p,g)\in TP} \mathrm{IoU}(p,g)}{2|TP| + |FP| + |FN|}, \tag{9}$$

Key Results and Ablative Studies

01

PanSt3R + LUDVIG achieves **state-of-the-art results** on the PanLift benchmark (ScanNet, Hypersim,Replica).

PanSt3R is orders of magnitude faster than previous methods. **The direct prediction method (4.5 minutes) is dramatically faster** than prior NeRF-based methods (e.g., Contrastive Lift at 420 minutes)

Method	Req. Poses	Hyper-sim	Rep-lica	Scan Net	Time (min)
DM-NeRF [57]	✓	51.6	44.1	41.7	~ 900
PNF [25]	✓	44.8	41.1	48.3	-
PanLift [50]	✓	60.1	57.9	58.9	~ 450
Contrastive Lift [2]	✓	62.3	59.1	62.3	~ 420
PLGS [62]	✓	62.4	57.8	58.7	~ 120
PCF-Lift [78]	✓	-	-	63.5	-
PanSt3R w/o QUBO	†	51.6	57.3	59.5	$\sim 4\ (+35^{\dagger})$
PanSt3R	†	56.5	**62.0**	65.7	$\sim 4.5\ (+35^{\dagger})$
PanSt3R + LUDVIG	✓	**66.3**	60.6	**67.5**	~ 40

02

On the ScanNet++ validation set (100 classes), PanSt3R **largely outperformed** competing NeRF-based methods by over 10% in PQ while also performed significantly better in predicting things (as indicated by PQth).

Table 3. Results on the ScanNet++ val set. We report result of our default model PanSt3R (full) and one trained only on ScanNet++. PanSt3R is compared with PanLift [50] and Contrastive Lift [2], utilizing Mask2Former [15] finetuned on ScanNet++.

Method	PQ	PQ_{th}	PQ_{st}	Time (min)
PanLift [50]	29.5	15.6	59.4	~ 500
Contrastive Lift [2]	28.4	14.8	56.3	~ 460
PanSt3R (ScanNet++)	46.7	43.2	55.8	~ 2.3
+ LUDVIG	54.8	52.4	62.4	~ 35
PanSt3R (full)	49.1	45.8	58.7	~ 2.3
+ LUDVIG	54.7	51.7	62.4	~ 35

03

Ablative studies indicated that combining 3D geometric features (MUSt3R) and 2D semantic features (DINOv2) is **essential and demonstrated a complementarity effect.**

QUBO resulted in a large boost in performance across datasets, highlighting the inadequacy of standard heuristic merging procedures for multi-view predictions

Table 4. Ablating the importance of 3D (MUSt3R [3]) and 2D (DINOv2 [38]) features on the ScanNet++ validation set. Results are shown for PanSt3R (224, ScanNet++)+LUDVIG.

features	PQ	PQ_{th}	PQ_{st}
3D + 2D	50.4	45.4	61.1
3D	46.4	40.7	58.8
2D	35.7	28.4	51.9

Table 5. Analyzing the effect of the QUBO merging strategy (Sec. 3.2). Results are reported for PanSt3R+LUDVIG.

QUBO	Hypersim	Replica	ScanNet	ScanNet++
x	58.1	60.8	60.2	50.8
✓	66.7	60.7	67.3	52.0

Key Innovations of PanSt3R

01

A unified, end-to-end framework that directly models **3D geometry and panoptic segmentation from unposed images without requiring camera parameters or depth input.**

02

QUBO (Quadratic Unconstrained Binary Optimisation) - a **novel and mathematically grounded mask prediction strategy to reconcile multi-view mask predictions,** which performs significantly better than heuristic merging procedures like those used in MaskFormer.

03

Introduces Two Novel View Synthesis Approaches:

1. Novel RGB view generation using 3DGS and a forward pass of PanSt3R on the rendered view.
2. Uplifting predicted panoptic segmentations to 3D and rendering **the segmentation on novel view using LUDVIG.**

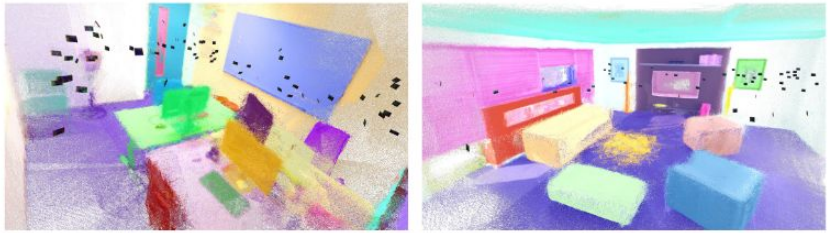


Figure 1. PanSt3R jointly predicts 3D geometry and panoptic segmentation of a scene in a **single forward pass**.

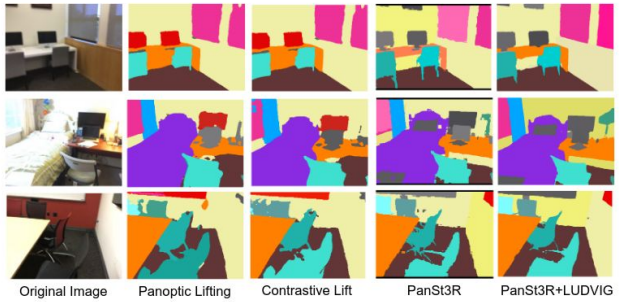


Figure 4. Qualitative comparison of novel-view panoptic segmentation on **ScanNet** [12] scenes. Colors and their nuances denote different classes and object instances respectively.

Limitations

- Dependency on Pre-trained Models
- Frozen Backbone Architecture - prevents end to end optimization benefits
- Insufficient Analysis of Failure Cases
- Evaluation limited to indoor scenes

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Conclusion

- Unified approach for joint 3D reconstruction and Panoptic Segmentation with SOTA results and faster speeds
- Joint prediction of 3D geometry and multi-view-consistent panoptic segmentation in a single forward pass eliminates the need for camera parameters, depth maps, and costly test-time optimization
- The successful fusion of DINOv2 and MUSt3R foundation models, combined with the principled QUBO-based mask merging strategy and inherent multi-view consistency through shared queries, establishes a new state of the art for Multi-view Consistent Panoptic Segmentation

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